

CUSTOMER PURCHASE BEHAVIOR ANALYSIS USING COVARIANCE AND CORRELATION

C03-AT1-POSTER PRPRESENTATION

STAT

DATA

AI

PY

Yedururi Anjani Priya
B . Tech AI & DS

Statistical Analysis of Customer Attributes and Purchasing Patterns

INTRODUCTION

Customer purchase data contains several measurable attributes. Covariance and correlation help identify how these variables change together and how strongly they are related.

PROBLEM

- Which customer variables are related?
- Does income relate to spending?
- Is purchase frequency associated with purchase amount?

APPROACH

- Clean the dataset
- Calculate covariance
- Calculate Pearson correlation
- Visualize and interpret

WHY?

- Find useful patterns
- Support segmentation
- Improve marketing decisions
- Enable data-driven planning

OBJECTIVES

- Analyze customer purchase behavior.
- Measure covariance between selected variables.
- Calculate Pearson correlation coefficients.
- Identify strong, weak, positive and negative relationships.
- Translate statistical findings into business insights.

METHOD PIPELINE



CONCLUSION

Covariance shows the direction in which two variables move together, while correlation measures the strength of their linear relationship. Together, they provide a simple statistical basis for understanding customer spending patterns and supporting business decisions.

DATASET

CUSTOMER VARIABLES

- Customer ID
- Age
- Gender
- Annual Income
- Customer Rating

PURCHASE VARIABLES

- Purchase Amount
- Purchase Frequency
- Product Category
- Discount / Promotion
- Total Spending

DATA PREPARATION

Missing values → Duplicates → Data types → Outliers → Final analysis dataset

STATISTICAL METHODS

COVARIANCE

$$\text{Cov}(X,Y) = \Sigma[(X_i - \bar{X})(Y_i - \bar{Y})] / (n - 1)$$

Positive → variables move together
Negative → variables move oppositely

CORRELATION

$$r = \text{Cov}(X,Y) / (\sigma_x \sigma_y)$$

+1 → strong positive
0 → weak / none
-1 → strong negative

VISUAL ANALYSIS

Income vs Purchase Amount



Correlation Matrix

	Age	Income	Amount	Freq.
Age	1.00	r=	r=	r=
Income	r=	1.00	r=	r=
Amount	r=	r=	1.00	r=
Freq.	Replace r= with final results.	r=	r=	1.00

RESULTS

RESULT 1

Variables: Income and Purchase Amount
Covariance: Positive
Interpretation: As customer income increases, purchase amount tends to increase.

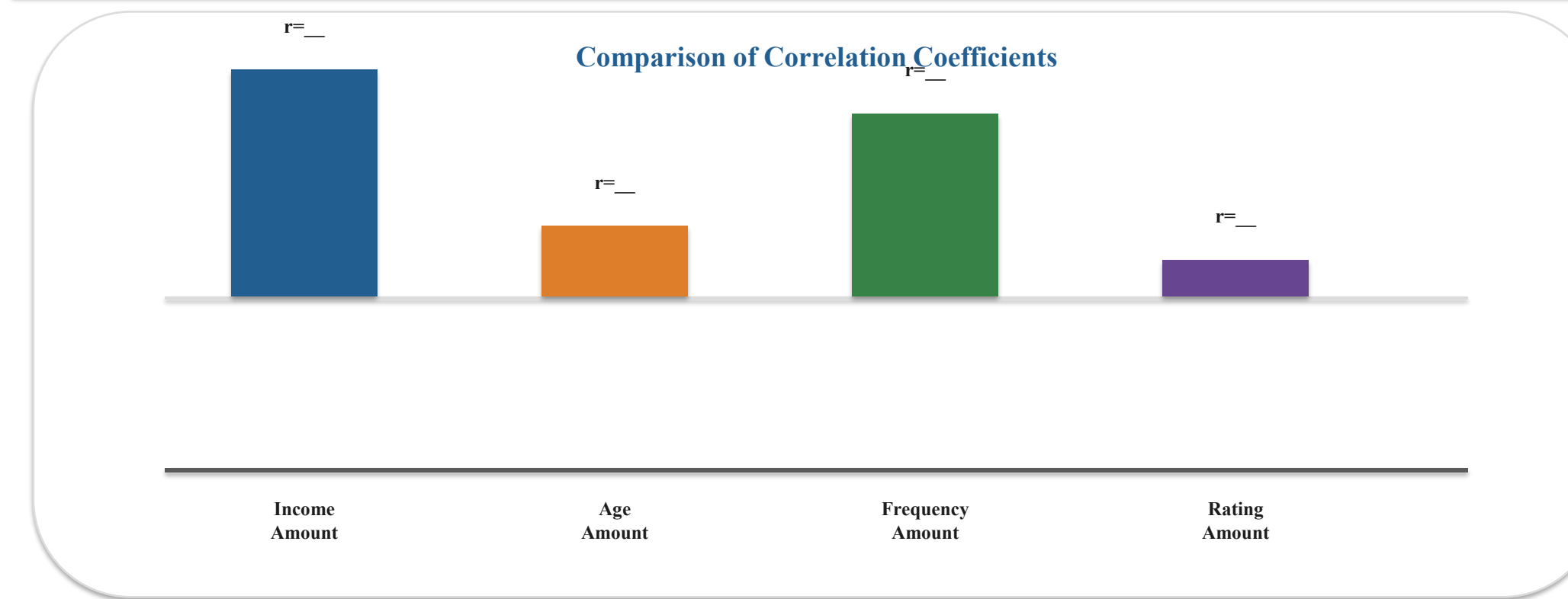
RESULT 2

Variables: Purchase Frequency and Total Spending
r = 0.72
Interpretation: Strong positive correlation.

RESULT 3

Variables: Age and Customer Rating
r = -0.18
Interpretation: Weak negative correlation.

CORRELATION RESULTS



KEY FINDINGS

- Strongest positive relationship: Purchase Amount (r = 0.85)
- Strongest negative relationship: Customer Rating (r = -0.18)
- Frequency–spending relationship: Strong Positive (r = 0.72)
- Weak / negligible relationship: Customer Rating – (r = 0.15)

BUSINESS INSIGHTS

- Identify high-value customers.
- Improve customer segmentation.
- Personalize promotions.
- Support marketing decisions.
- Guide data-driven planning.